



## **Módulo 2 - Práctica Final**

Hanuel José Martínez Guzmán

JA Dominicana

Administración de bases de datos

Jhoan Báez

Santo Domingo, 21 de Junio de 2026

## Creación y Manipulación de la Base de Datos "Practica4"

### Ejercicio 1:

**CREATE DATABASE** Practica\_Final;

| Name         | Value                          |
|--------------|--------------------------------|
| Updated Rows | 1                              |
| Execute time | 0.001s                         |
| Start time   | Sun Jun 21 11:41:55 AST 2026   |
| Finish time  | Sun Jun 21 11:41:55 AST 2026   |
| Query        | CREATE DATABASE Practica_Final |

### Ejercicio 2:

**USE** Practica\_Final;

| Name         | Value                        |
|--------------|------------------------------|
| Updated Rows | 0                            |
| Execute time | 0.001s                       |
| Start time   | Sun Jun 21 11:43:24 AST 2026 |
| Finish time  | Sun Jun 21 11:43:24 AST 2026 |
| Query        | USE Practica_Final           |

**Ejercicio 3:**

```
CREATE TABLE Pais(  
    id_pais INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,  
    nombre VARCHAR(60) UNIQUE NOT NULL,  
    estado CHAR(1) NOT NULL,  
    CONSTRAINT check_pais_estado CHECK(estado IN("A", "I"))  
);
```

| Name         | Value                                                                                                                                                                                                                                              |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                  |
| Execute time | 0.03s                                                                                                                                                                                                                                              |
| Start time   | Sun Jun 21 12:29:31 AST 2026                                                                                                                                                                                                                       |
| Finish time  | Sun Jun 21 12:29:31 AST 2026                                                                                                                                                                                                                       |
| Query        | <pre>CREATE TABLE Pais(<br/>    id_pais INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,<br/>    nombre VARCHAR(60) UNIQUE NOT NULL,<br/>    estado CHAR(1) NOT NULL,<br/>    CONSTRAINT check_pais_estado CHECK(estado IN("A", "I"))<br/>)</pre> |

```
CREATE TABLE Departamento(  
    id_departamento INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,  
    nombre VARCHAR(60) UNIQUE NOT NULL,  
    estado CHAR(1) NOT NULL,  
    CONSTRAINT check_departamento_estado CHECK(estado IN("A", "I"))  
);
```

| Name         | Value                                                                                                                                                                                                                                                  |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                      |
| Execute time | 0.022s                                                                                                                                                                                                                                                 |
| Start time   | Sun Jun 21 12:30:14 AST 2026                                                                                                                                                                                                                           |
| Finish time  | Sun Jun 21 12:30:14 AST 2026                                                                                                                                                                                                                           |
| Query        | <pre>CREATE TABLE Departamento(     id_departamento INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,     nombre VARCHAR(60) UNIQUE NOT NULL,     estado CHAR(1) NOT NULL,     CONSTRAINT check_departamento_estado CHECK(estado IN("A", "I")) )</pre> |

**CREATE TABLE Empleado(**

```

    id_empleado INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,
    nombre VARCHAR(60) NOT NULL,
    apellido VARCHAR(60) NOT NULL,
    cedula CHAR(11) UNIQUE NOT NULL,
    telefono CHAR(10) NOT NULL,
    fecha_nacimiento DATE NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    id_supervisor INT UNSIGNED,
    genero CHAR(1) NOT NULL,
    CONSTRAINT check_empleado_genero CHECK(genero IN("M", "F")),
    id_pais INT UNSIGNED NOT NULL,
    condicion CHAR(1) NOT NULL,
    CONSTRAINT check_empleado_condicion CHECK(condicion IN("A", "L", "V",
"S")),
    estado_civil CHAR(1) NOT NULL,
```

```
        CONSTRAINT check_empleado_estado_civil CHECK(estado_civil IN("C", "S",  
"D", "V")),  
        fecha_ingreso DATETIME NOT NULL DEFAULT NOW(),  
        fecha_salida DATE,  
        salario DECIMAL(7, 2) NOT NULL,  
        id_departamento INT UNSIGNED NOT NULL,  
        FOREIGN KEY (id_supervisor)  
            REFERENCES Empleado(id_empleado),  
        FOREIGN KEY (id_pais)  
            REFERENCES Pais(id_pais),  
        FOREIGN KEY (id_departamento)  
            REFERENCES Departamento(id_departamento)  
    );
```

| Name         | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Execute time | 0.04s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Start time   | Sun Jun 21 14:13:13 AST 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Finish time  | Sun Jun 21 14:13:13 AST 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Query        | <pre> CREATE TABLE Empleado(     id_empleado INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,     nombre VARCHAR(60) NOT NULL,     apellido VARCHAR(60) NOT NULL,     cedula CHAR(11) UNIQUE NOT NULL,     telefono CHAR(10) NOT NULL,     fecha_nacimiento DATE NOT NULL,     email VARCHAR(100) UNIQUE NOT NULL,     id_supervisor INT UNSIGNED,     genero CHAR(1) NOT NULL,     CONSTRAINT check_empleado_genero CHECK(genero IN("M", "F")),     id_pais INT UNSIGNED NOT NULL,     condicion CHAR(1) NOT NULL,     CONSTRAINT check_empleado_condicion CHECK(condicion IN("A", "L", "V", "S")),     estado_civil CHAR(1) NOT NULL,     CONSTRAINT check_empleado_estado_civil CHECK(estado_civil IN("C", "S", "D", "V")),     fecha_ingreso DATETIME NOT NULL DEFAULT NOW(),     fecha_salida DATE,     salario DECIMAL(7, 2) NOT NULL,     id_departamento INT UNSIGNED NOT NULL,     FOREIGN KEY (id_supervisor)         REFERENCES Empleado(id_empleado),     FOREIGN KEY (id_pais)         REFERENCES Pais(id_pais),     FOREIGN KEY (id_departamento)         REFERENCES Departamento(id_departamento) ) </pre> |

## CREATE TABLE Usuario(

**id\_usuario INT UNSIGNED PRIMARY KEY AUTO\_INCREMENT NOT NULL,**

**id\_empleado INT UNSIGNED NOT NULL,**

**usuario VARCHAR(60) UNIQUE NOT NULL,**

**pwd VARBINARY(128) NOT NULL,**

**FOREIGN KEY (id\_empleado)**

**REFERENCES Empleado(id\_empleado)**

);

| Name         | Value                                                                                                                                                                                                                                                                                       |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                                                           |
| Execute time | 0.027s                                                                                                                                                                                                                                                                                      |
| Start time   | Sun Jun 21 14:14:38 AST 2026                                                                                                                                                                                                                                                                |
| Finish time  | Sun Jun 21 14:14:38 AST 2026                                                                                                                                                                                                                                                                |
| Query        | <pre>CREATE TABLE Usuario(     id_usuario INT UNSIGNED PRIMARY KEY AUTO_INCREMENT NOT NULL,     id_empleado INT UNSIGNED NOT NULL,     usuario VARCHAR(60) UNIQUE NOT NULL,     pwd VARBINARY(128) NOT NULL,     FOREIGN KEY (id_empleado)         REFERENCES Empleado(id_empleado) )</pre> |

#### Ejercicio 4:

**DESCRIBE Pais;**

|   | AZ Field ▼ | AZ Type ▼        | AZ Null ▼ | AZ Key ▼ | AZ Default ▼ | AZ Extra ▼     |
|---|------------|------------------|-----------|----------|--------------|----------------|
| 1 | id_pais    | int(10) unsigned | NO        | PRI      | [NULL]       | auto_increment |
| 2 | nombre     | varchar(60)      | NO        | UNI      | [NULL]       |                |
| 3 | estado     | char(1)          | NO        |          | [NULL]       |                |
|   |            |                  |           |          |              |                |

#### Ejercicio 5:

**DESCRIBE** Departamento;

|   | AZ Field ▼      | AZ Type ▼        | AZ Null ▼ | AZ Key ▼ | AZ Default ▼ | AZ Extra ▼     |
|---|-----------------|------------------|-----------|----------|--------------|----------------|
| 1 | id_departamento | int(10) unsigned | NO        | PRI      | [NULL]       | auto_increment |
| 2 | nombre          | varchar(60)      | NO        | UNI      | [NULL]       |                |
| 3 | estado          | char(1)          | NO        |          | [NULL]       |                |
|   |                 |                  |           |          |              |                |
|   |                 |                  |           |          |              |                |

**Ejercicio 6:**

**DESCRIBE** Empleado;

|    | AZ Field ▼       | AZ Type ▼        | AZ Null ▼ | AZ Key ▼ | AZ Default ▼        | AZ Extra ▼     |
|----|------------------|------------------|-----------|----------|---------------------|----------------|
| 1  | id_empleado      | int(10) unsigned | NO        | PRI      | [NULL]              | auto_increment |
| 2  | nombre           | varchar(60)      | NO        |          | [NULL]              |                |
| 3  | apellido         | varchar(60)      | NO        |          | [NULL]              |                |
| 4  | cedula           | char(11)         | NO        | UNI      | [NULL]              |                |
| 5  | telefono         | char(10)         | NO        |          | [NULL]              |                |
| 6  | fecha_nacimiento | date             | NO        |          | [NULL]              |                |
| 7  | email            | varchar(100)     | NO        | UNI      | [NULL]              |                |
| 8  | id_supervisor    | int(10) unsigned | YES       | MUL      | [NULL]              |                |
| 9  | genero           | char(1)          | NO        |          | [NULL]              |                |
| 10 | id_pais          | int(10) unsigned | NO        | MUL      | [NULL]              |                |
| 11 | condicion        | char(1)          | NO        |          | [NULL]              |                |
| 12 | estado_civil     | char(1)          | NO        |          | [NULL]              |                |
| 13 | fecha_ingreso    | datetime         | NO        |          | current_timestamp() |                |
| 14 | fecha_salida     | date             | YES       |          | [NULL]              |                |
| 15 | salario          | decimal(7,2)     | NO        |          | [NULL]              |                |
| 16 | id_departamento  | int(10) unsigned | NO        | MUL      | [NULL]              |                |
|    |                  |                  |           |          |                     |                |
|    |                  |                  |           |          |                     |                |

**Ejercicio 7:**

**DESCRIBE** Usuario;



|   | AZ Field    | AZ Type          | AZ Null | AZ Key | AZ Default | AZ Extra       |
|---|-------------|------------------|---------|--------|------------|----------------|
| 1 | id_usuario  | int(10) unsigned | NO      | PRI    | [NULL]     | auto_increment |
| 2 | id_empleado | int(10) unsigned | NO      | MUL    | [NULL]     |                |
| 3 | usuario     | varchar(60)      | NO      | UNI    | [NULL]     |                |
| 4 | pwd         | varbinary(128)   | NO      |        | [NULL]     |                |

### Ejercicio 8:

**INSERT INTO** Pais (nombre, estado)

**VALUES**

('Republica Dominicana', 'A'),

('Mexico', 'A'),

('Colombia', 'A');

| Name         | Value                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 3                                                                                                                       |
| Execute time | 0.009s                                                                                                                  |
| Start time   | Sun Jun 21 12:37:19 AST 2026                                                                                            |
| Finish time  | Sun Jun 21 12:37:19 AST 2026                                                                                            |
| Query        | <pre> INSERT INTO Pais (nombre, estado) VALUES ('Republica Dominicana', 'A'), ('Mexico', 'A'), ('Colombia', 'A') </pre> |

### Ejercicio 9:

**INSERT INTO** Departamento (nombre, estado)

**VALUES**

('Recursos Humanos', 'A'),

('Tecnologia', 'A'),

('Contabilidad', 'A');

| Name         | Value                                                                                                                              |
|--------------|------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 3                                                                                                                                  |
| Execute time | 0.003s                                                                                                                             |
| Start time   | Sun Jun 21 12:38:28 AST 2026                                                                                                       |
| Finish time  | Sun Jun 21 12:38:28 AST 2026                                                                                                       |
| Query        | INSERT INTO Departamento (nombre, estado)<br>VALUES<br>('Recursos Humanos', 'A'),<br>('Tecnologia', 'A'),<br>('Contabilidad', 'A') |

**Ejercicio 10:**

**INSERT INTO** Empleado (

nombre,

apellido,

cedula,

telefono,

fecha\_nacimiento,

email,

id\_supervisor,

```
genero,
id_pais,
condicion,
estado_civil,
salario,
id_departamento
)
VALUES
(
  'Juan',
  'Perez',
  '00112345678',
  '8095551111',
  '1990-05-15',
  'juan.perez@empresa.com',
  NULL,
  'M',
  1,
  'A',
  'S',
  35000.00,
  1
),
(
  'Maria',
  'Gomez',
```

'00212345678',  
'8095552222',  
'1992-08-20',  
'maria.gomez@empresa.com',  
1,  
'F',  
2,  
'A',  
'C',  
45000.00,  
2  
)  
(  
'Carlos',  
'Rodriguez',  
'00312345678',  
'8095553333',  
'1988-11-10',  
'carlos.rodriguez@empresa.com',  
1,  
'M',  
3,  
'V',  
'D',  
40000.00,  
3

);

| Name         | Value                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 3                                                                                                                                                                                                                                                                                                                                                                                             |
| Execute time | 0.004s                                                                                                                                                                                                                                                                                                                                                                                        |
| Start time   | Sun Jun 21 14:20:42 AST 2026                                                                                                                                                                                                                                                                                                                                                                  |
| Finish time  | Sun Jun 21 14:20:42 AST 2026                                                                                                                                                                                                                                                                                                                                                                  |
| Query        | <pre>INSERT INTO Empleado (<br/>    nombre,<br/>    apellido,<br/>    cedula,<br/>    telefono,<br/>    fecha_nacimiento,<br/>    email,<br/>    id_supervisor,<br/>    genero,<br/>    id_pais,<br/>    condicion,<br/>    estado_civil,<br/>    salario,<br/>    id_departamento<br/>)<br/>VALUES<br/>(<br/>    'Juan',<br/>    'Perez',<br/>    '00112345678',<br/>    '8095551111',</pre> |

### Ejercicio 11:

**INSERT INTO** Usuario (id\_empleado, usuario, pwd)

**VALUES**

```
(  
    1,  
    'jperez',  
    SHA2('Juan123', 512)  
)  
  
(  
    2,  
    'mgomez',  
    SHA2('Maria123', 512)  
)  
  
(  
    3,  
    'crodriguez',  
    SHA2('Carlos123', 512)  
);
```

| Name         | Value                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 3                                                                                                                                                                                                                                |
| Execute time | 0.004s                                                                                                                                                                                                                           |
| Start time   | Sun Jun 21 12:42:09 AST 2026                                                                                                                                                                                                     |
| Finish time  | Sun Jun 21 12:42:09 AST 2026                                                                                                                                                                                                     |
| Query        | <pre>INSERT INTO Usuario (id_empleado, usuario, pwd) VALUES (     1,     'jperez',     SHA2('Juan123', 512) ), (     2,     'mgomez',     SHA2('Maria123', 512) ), (     3,     'crodriguez',     SHA2('Carlos123', 512) )</pre> |

### Ejercicio 12:

**DELIMITER //**

**CREATE FUNCTION** f\_calcular\_edad(fecha\_nacimiento **DATE**)

**RETURNS INT**

```

BEGIN

    DECLARE age INT;

    SET age = YEAR(NOW()) - YEAR(fecha_nacimiento);

    IF (MONTH(NOW()) > MONTH(fecha_nacimiento)) OR (MONTH(NOW()) =
MONTH(fecha_nacimiento) AND DAY(NOW()) > DAY(fecha_nacimiento)) THEN

        SET age = age - 1;

    END IF;

    RETURN age;

END

//

```

| Name         | Value                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                                                                                                                                       |
| Execute time | 0.009s                                                                                                                                                                                                                                                                                                                                                                  |
| Start time   | Sun Jun 21 12:44:13 AST 2026                                                                                                                                                                                                                                                                                                                                            |
| Finish time  | Sun Jun 21 12:44:13 AST 2026                                                                                                                                                                                                                                                                                                                                            |
| Query        | <pre> CREATE FUNCTION f_calcular_edad(fecha_nacimiento DATE) RETURNS INT BEGIN     DECLARE age INT;     SET age = YEAR(NOW()) - YEAR(fecha_nacimiento);     IF (MONTH(NOW()) &gt; MONTH(fecha_nacimiento)) OR (MONTH(NOW()) = MONTH(fecha_nacimiento) AND DAY(NOW()) &gt; DAY(fecha_nacimiento)) THEN         SET age = age - 1;     END IF;     RETURN age; END </pre> |

### Ejercicio 13:

**SELECT**

e.nombre,



e.apellido,  
 e.cedula,  
 e.telefono,  
 f\_calcular\_edad(e.fecha\_nacimiento) AS edad,  
 e.email,  
 e2.nombre **AS** supervisor,  
 e.genero,  
 p.nombre **AS** pais,  
 e.condicion,  
 e.estado\_civil,  
 e.salario,  
 d.nombre **AS** departamento  
**FROM** Empleado e  
**LEFT JOIN** Departamento d  
**ON** e.id\_departamento = d.id\_departamento  
**LEFT JOIN** Pais p  
**ON** e.id\_pais = p.id\_pais  
**LEFT JOIN** Empleado e2  
**ON** e.id\_supervisor = e2.id\_employado;

|   | Az nombre | Az apellido | Az cedula   | Az telefono | id edad | Az email                     | Az supervisor | Az genero | Az pais              | Az condicion | Az estado_civil | id salario | Az departamento  |
|---|-----------|-------------|-------------|-------------|---------|------------------------------|---------------|-----------|----------------------|--------------|-----------------|------------|------------------|
| 1 | Juan      | Perez       | 0012345678  | 8095551111  | 35      | juan.perez@empresa.com       | [NULL]        | M         | Republica Dominicana | A            | S               | 35,000     | Recursos Humanos |
| 2 | Maria     | Gomez       | 00212345678 | 8095552222  | 34      | maria.gomez@empresa.com      | Juan          | F         | Mexico               | A            | C               | 45,000     | Tecnologia       |
| 3 | Carlos    | Rodriguez   | 00312345678 | 8095553333  | 38      | carlos.rodriguez@empresa.com | Juan          | M         | Colombia             | V            | D               | 40,000     | Contabilidad     |

## Ejercicio 14:

**CREATE VIEW v\_empleado AS**

**SELECT**

```
    e.nombre,  
  
    e.apellido,  
  
    e.cedula,  
  
    e.telefono,  
  
    f_calcular_edad(e.fecha_nacimiento) AS edad,  
  
    e.email,  
  
    e2.nombre AS supervisor,  
  
    e.genero,  
  
    p.nombre AS pais,  
  
    e.condicion,  
  
    e.estado_civil,  
  
    e.salario,  
  
    d.nombre AS departamento  
  
    FROM Empleado e  
  
    LEFT JOIN Departamento d  
        ON e.id_departamento = d.id_departamento  
  
    LEFT JOIN Pais p  
        ON e.id_pais = p.id_pais  
  
    LEFT JOIN Empleado e2  
        ON e.id_supervisor = e2.id_empleado;
```

| Name         | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Updated Rows | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Execute time | 0.016s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Start time   | Sun Jun 21 14:27:52 AST 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Finish time  | Sun Jun 21 14:27:52 AST 2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Query        | <pre>CREATE VIEW v_empleado AS     SELECT         e.nombre,         e.apellido,         e.cedula,         e.telefono,         f_calcular_edad(e.fecha_nacimiento) AS edad,         e.email,         e2.nombre AS supervisor,         e.genero,         p.nombre AS pais,         e.condicion,         e.estado_civil,         e.salario,         d.nombre AS departamento     FROM Empleado e     LEFT JOIN Departamento d         ON e.id_departamento = d.id_departamento     LEFT JOIN Pais p         ON e.id_pais = p.id_pais     LEFT JOIN Empleado e2         ON e.id_supervisor = e2.id_empleado</pre> |